FEAT-ZD-48600 · REQ-05 Product Requirement Preparation · Approved PRD

Requirement agent · item feat.prd · source hobasa-requirement-agent/outputs/FEAT-ZD-48600/feat.prd.md · approved by PM sim.pm

feat.prd: FEAT-ZD-48600

Product Requirement Document for FEAT-ZD-48600 (ZD-48600, "Show which overdue invoices belong to vendors that usually pay late"). This is a draft for the REQ-05 gate (PM).

This PRD carries forward the decisions in the approved BRD (feat.brd). It adds no scope. It turns the BRD scope (S-1 to S-7) and business rules (BR-1 to BR-16) into user stories, numbered acceptance criteria in EARS form and non-functional requirements, and ends with a validation check. Statements marked "agent judgement" are this agent's own reading; no input states them as fact. PRD-level assumptions (P-A) and questions (PQ) are new in this step and need confirmation at the gate.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief; feat.brd
Intake	ZD-48600, Zendesk ticket (sample, synthetic), channel Customer Feedback, received 2026-10-24	feat.brd; file:hobasa-agentic-sdlc/requirements/ZD-48600.md
Source link (GitHub)	`hobasa-agentic-sdlc/requirements/ZD-48600.md`	feat.brd
Trigger	BRD approved	record:brief
Fitment	"Yes", linked roadmap item R-AP-12 ("give AP managers earlier visibility of overdue payables")	feat.brd
Category / qualifier	New Feature / Not Technical Only	feat.brd
Risk tier (advisory)	T2. "The risk label is advisory only."	feat.brd; file:hobasa-agentic-sdlc/requirements/ZD-48600.md
Repositories	ap_pipeline, ap_forecast, ap_aging, ap_portal	feat.brd; file:hobasa-agentic-sdlc/requirements/ZD-48600.md
Owner / gate of this step	PM / PM	record:brief
Template	PRD template: summary, user stories, numbered acceptance criteria (EARS: When <trigger>, the system shall <response>), non-functional requirements, open questions	org.templates

1. Summary

Item	Statement	Source
Problem	"AP managers want to see which overdue invoices belong to vendors that usually pay late, so they can chase them first."	feat.brd (section 1)
What the product will do	Build each vendor's payment-delay history from a new optional `paid_date` in the vendor feed, give each vendor a rule-based risk level (`high`, `medium`, `low`) from its predicted delay, list unpaid invoices more than 30 days past due with that risk level, and show the list as a table in the portal.	feat.brd (sections 2, 4)
Flow	feed -> delays -> risk -> overdue list -> portal table, proven end to end.	feat.brd (S-7, SM-1)

Item	Statement	Source
Users	AP managers. Finance teams benefit indirectly (agent judgement carried from the BRD).	feat.brd (section 3)
Business meaning	Carried from the BRD as an assumption: the risk level helps AP managers decide which overdue payables to act on first (B-A1, still open as OQ-B1).	feat.brd (B-A1, OQ-B1)
Architecture rule	"Keep every existing signature; add new functions." Existing behaviour changes only in two additive ways: the optional `risk` argument on `build_explanation_request` (CR-4) and feed ingestion accepting the optional `paid_date` (CR-1).	feat.brd (section 4, SM-5)
Not in this change	Finance posting, authentication, tenancy and payment execution; acting on the risk level automatically; an aging export (R-AP-14); a change of LLM provider or model; using an LLM to set the risk level; the work of ZD-48501 and ZD-48211; the regulatory "Late" rule (REG-LPR-2026-4.2).	feat.brd (OS-1 to OS-8)

2. User stories

Stories follow the story template's "As a / I want / So that" form. They are product-level stories; the backlog stories in NEW state are written later, in REQ-08. The "So that" of US-1 comes from the BRD (B-A1). The "So that" clauses of US-2 to US-5, and the choice of the PM as the actor in US-5, are agent judgement: the BRD states the rules and scope behind them but not these purposes.

ID	As a	I want	So that	BRD scope	Source
US-1	AP manager	to see, in the portal, the unpaid invoices that are more than 30 days past due, each with its vendor's risk level, sorted by days overdue (most overdue first)	I can decide which overdue payables to act on first (B-A1)	S-5, S-6	feat.brd (section 2, S-5, S-6, B-A1)
US-2	AP manager	each vendor's payment-delay history to be built from the date each invoice was actually paid	the risk level reflects how late the vendor's invoices have actually been settled	S-1, S-2	feat.brd (S-1, S-2, BR-3)
US-3	AP manager	each vendor to get a risk level from a fixed, explicit rule on its predicted delay, with no LLM involved	I can trust and trace why a vendor is marked high, medium or low	S-3	feat.brd (S-3, BR-4, BR-6, section 2 product-promise link)
US-4	AP manager	the payment-delay explanation to state the vendor's risk level when one is given	the explanation and the table tell the same story	S-4	feat.brd (S-4, BR-15)
US-5	PM	the whole flow to be proven end to end, from the feed to the portal table	I know the four repositories work together before release	S-7	feat.brd (S-7, SM-1); record:brief

3. Acceptance criteria (EARS)

Each criterion uses the EARS form "When <trigger>, the system shall <response>" (org.templates). "Days" are calendar days after the due date (BR-9, B-A2).

3.1 Vendor feed `paid_date` (ap_pipeline, S-1, US-2)

#	Acceptance criterion	Traces to	Source
AC-1	When a vendor feed row has an empty `paid_date`, the system shall accept the row and set `Invoice.paid_date` to None.	BR-1, S-1	feat.brd (BR-1, S-1); file:hobasa-agentic-sdlc/requirements/ZD-48600.md (org.data_model: "paid_date (date or None)")
AC-2	When a vendor feed row that is marked paid has a `paid_date` that is a valid ISO date, the system shall map it to `Invoice.paid_date`. (A row that is unpaid but has a valid `paid_date` is not covered here; its handling is open as OQ-B6.)	BR-1, S-1	feat.brd (BR-1, S-1, OQ-B6)
AC-3	When a vendor feed row has a `paid_date` that is not empty and is not a valid ISO date, the system shall reject the row with the reason `'paid_date'`.	BR-2, SM-4	feat.brd (BR-2, SM-4); assumption P-A3

3.2 Payment-delay history (ap_pipeline, S-2, US-2)

#	Acceptance criterion	Traces to	Source
AC-4	When `delays_by_vendor(records)` is called, the system shall return `{vendor: [delay_days, ...]}` built only from records that are paid and have a `paid_date`, where `delay_days = (paid_date - due_date)` in days, with each vendor's values in feed order.	BR-3, S-2	feat.brd (BR-3, S-2)
AC-5	When a record is marked paid but has no `paid_date`, the system shall leave it out of the delay history and shall not reject it.	B-A5	feat.brd (B-A5)
AC-6	When a paid record's `paid_date` is before its `due_date`, the system shall keep the negative `delay_days` as computed, with no floor.	B-A4	feat.brd (B-A4)

3.3 Vendor risk level and explanation (ap_forecast, S-3, S-4, US-3, US-4)

#	Acceptance criterion	Traces to	Source
AC-7	When `vendor_risk(history)` is called and `predict_delay_days(history) > 30`, the system shall return `'high'`.	BR-4	feat.brd (BR-4)

#	Acceptance criterion	Traces to	Source
AC-8	When `vendor_risk(history)` is called and `predict_delay_days(history)` is greater than 10 and not greater than 30 (including exactly 30), the system shall return `'medium'`.	BR-4, BR-5, SM-2	feat.brd (BR-4, BR-5, SM-2)
AC-9	When `vendor_risk(history)` is called and `predict_delay_days(history)` is not greater than 10 (including exactly 10), the system shall return `'low'`.	BR-4, BR-5, SM-2	feat.brd (BR-4, BR-5, SM-2)
AC-10	When `vendor_risk` computes a risk level, the system shall apply the rule on the predicted delay only and shall not call an LLM.	BR-6, OS-5	feat.brd (BR-6, OS-5)
AC-11	When `build_explanation_request` is called with a `risk` argument, the system shall build a prompt that ends with `'Risk level: <risk>.'`.	BR-15, S-4	feat.brd (BR-15, S-4)
AC-12	When `build_explanation_request` is called without a `risk` argument, the system shall build the same prompt as today.	B-A6, SM-5	feat.brd (B-A6, SM-5)

3.4 Overdue list with risk (ap_aging, S-5, US-1)

#	Acceptance criterion	Traces to	Source
AC-13	When `overdue_with_risk(invoices, as_of, risk_by_vendor)` is called, the system shall include an invoice only when it is unpaid and more than 30 days past due as of `as_of`, counting calendar days after the due date.	BR-7, BR-9, S-5	feat.brd (BR-7, BR-9, S-5); assumption P-A1
AC-14	When an unpaid invoice is exactly 30 days past due as of `as_of`, the system shall not include it in the overdue list.	BR-7, SM-2	feat.brd (BR-7, SM-2); record:memory (L-007)
AC-15	When an invoice is paid, the system shall exclude it from the overdue list, whatever its days past due.	BR-8	feat.brd (BR-8); record:memory (D-2026-050)
AC-16	When the overdue list is returned, the system shall give each row the keys `id`, `vendor`, `days`, `amount` and `risk`, and shall sort the rows by `days` descending, then by `id`.	BR-10	feat.brd (BR-10); assumption P-A2
AC-17	When an invoice's vendor has no entry in `risk_by_vendor`, the system shall set that row's `risk` to `'unknown'`.	BR-11	feat.brd (BR-11)

#	Acceptance criterion	Traces to	Source
AC-18	When the system sets a row's `amount`, it shall use the Invoice `amount` as a Decimal with 2 dp; when the list is passed from ap_aging to ap_portal as JSON, the system shall carry the amount as a string with two decimals and dates as ISO strings.	BR-12, B-A7, SM-3	feat.brd (BR-12, B-A7, SM-3)

3.5 Portal table (ap_portal, S-6, US-1)

#	Acceptance criterion	Traces to	Source
AC-19	When `renderOverdueRisk(rows)` is called with a non-empty list of rows, the system shall render `<table class="overdue-risk">` with one `<tr class="risk-RISK">` per row, showing id, vendor, days and amount to two decimals under the columns Invoice, Vendor, Days overdue and Amount.	BR-13, BR-14, S-6	feat.brd (BR-13, BR-14, S-6)
AC-20	When a row is rendered, the system shall set its class to `risk-high`, `risk-medium`, `risk-low` or `risk-unknown` to match the row's `risk` value.	BR-14	feat.brd (BR-14)
AC-21	When `renderOverdueRisk(rows)` is called with no rows, the system shall return an empty string.	BR-13	feat.brd (BR-13); file:hobasa-agentic-sdlc/requirements/ZD-48600.md (org.design_library: "render nothing when there are no rows")

3.6 End to end (cross-repository, S-7, US-5)

#	Acceptance criterion	Traces to	Source
AC-22	When the end-to-end test runs on a vendor feed, the system shall carry the data through feed -> delays -> risk -> overdue list -> portal table, and the rendered table shall match the overdue list built from that feed.	S-7, SM-1	feat.brd (S-7, SM-1); assumption P-A4
AC-23	When the end-to-end test runs, the system shall pass each vendor's `delay_days` list from `delays_by_vendor` to `vendor_risk` as its `history`, and shall pass the resulting vendor-to-risk map to `overdue_with_risk` as `risk_by_vendor`.	B-A3	feat.brd (B-A3)

4. Non-functional requirements

ID	Requirement	Traces to	Source
NFR-1	Monetary amounts use Decimal, never float; Decimal arithmetic is kept and amounts are never converted to float.	BR-12, SM-3	feat.brd (BR-12, SM-3); record:memory (D-2026-044, L-011)
NFR-2	JSON between services carries amounts as strings with two decimals and dates as ISO strings.	BR-12	feat.brd (BR-12); file:hobasa-agentic-sdlc/requirements/ZD-48600.md (org.architecture)
NFR-3	ap_portal shows the amount to two decimals without converting it to a binary floating-point number. Agent judgement: D-2026-044 applies to ap_portal too. How ap_portal does this is left to the TRD (REQ-07); OQ-B10 stays open for confirmation.	SM-3, OQ-B10	feat.brd (SM-3, OQ-B10); record:memory (D-2026-044)
NFR-4	Backward compatibility: "Keep every existing signature; add new functions." Only the two additive changes CR-1 and CR-4 touch existing behaviour; callers that do not pass `risk` see no change (AC-12).	SM-5, OS-6	feat.brd (section 4, SM-5, OS-6)
NFR-5	Auditability: the risk level comes from an explicit rule (no LLM), and rejected feed rows carry a named reason (`'paid_date'`).	BR-2, BR-6	feat.brd (section 2 product-promise link, BR-2, BR-6)
NFR-6	The risk label is advisory only. No risk value shall trigger an automatic action, posting or payment.	BR-16, OS-1, OS-2	feat.brd (BR-16, OS-1, OS-2)
NFR-7	Security and tenancy: no finance posting, authentication, tenancy or payment execution in this change (risk tier T2). Whether `renderOverdueRisk` must HTML-escape vendor names and ids is open (OQ-B8).	OS-1, OQ-B8	feat.brd (OS-1, OQ-B8); file:hobasa-agentic-sdlc/requirements/ZD-48600.md (org.risk_map)
NFR-8	Testability: boundary cases are tested explicitly: an unpaid invoice exactly 30 days past due (not listed, AC-14), and predicted delays of exactly 30 (`'medium'`, AC-8) and exactly 10 (`'low'`, AC-9).	SM-2	feat.brd (SM-2); record:memory (L-007)
NFR-9	Feed schedule unchanged: daily 02:00 Asia/Kolkata.	S-1	feat.brd (S-1: "The schedule is unchanged"); file:hobasa-agentic-sdlc/requirements/ZD-48600.md (org.customer_mappings: "Schedule: daily 02:00 Asia/Kolkata (unchanged)")
NFR-10	Terminology: the portal and later documents keep "overdue" (more than 30 days past due) distinct from the regulatory "Late" label (more than 60 days, REG-LPR-2026-4.2). Carried as a recommendation; OQ-B2 stays open.	OS-8, OQ-B2	feat.brd (OS-8, OQ-B2)

ID	Requirement	Traces to	Source
NFR-11	Performance, data volume and availability targets: not given in the inputs (PQ-4).	—	feat.brd; file:hobasa-agentic-sdlc/requirements/ZD-48600.md

Background, not a requirement: org.architecture describes ap_pipeline, ap_forecast and ap_aging as Python 3.9 and ap_portal as Node.js 22 CommonJS (file:hobasa-agentic-sdlc/requirements/ZD-48600.md). No input sets a requirement on runtimes, so this PRD sets none; runtime matters are for the TRD (REQ-07).

5. Assumptions

5.1 Carried from the approved BRD

ID	Assumption	Used in	Source
B-A1	The risk level helps AP managers decide which overdue payables to act on first.	Summary, US-1	feat.brd
B-A2	"Days past due" and `delay_days` are calendar days, following D-2026-031.	AC-4, AC-13	feat.brd; record:memory (D-2026-031)
B-A3	`history` for `vendor_risk` is one vendor's `delay_days` list from `delays_by_vendor`; `risk_by_vendor` maps each vendor to its `vendor_risk` result.	AC-23	feat.brd
B-A4	A negative `delay_days` is kept as computed.	AC-6	feat.brd
B-A5	A paid record with no `paid_date` is left out of the delay history, not rejected.	AC-5	feat.brd
B-A6	Without `risk`, `build_explanation_request` builds today's prompt.	AC-12	feat.brd
B-A7	The overdue-list `amount` is the Invoice `amount` (Decimal, 2 dp) and travels to ap_portal as a two-decimal string.	AC-18	feat.brd

5.2 New in this PRD (please confirm at the gate)

ID	Assumption	Used in	Source
P-A1	`days` in an overdue row is the number of calendar days from `due_date` to `as_of` (agent judgement from BR-9 and the `as_of` argument).	AC-13, AC-16	feat.brd (BR-9, S-5)
P-A2	"Then by id" means ascending order of `id`, compared as text, because the data model types `id` as a string (agent judgement; the ticket does not state the direction).	AC-16	feat.brd (BR-10); file:hobasa-agentic-sdlc/requirements/ZD-48600.md (org.data_model: "id (str)")
P-A3	An "invalid" `paid_date` is a non-empty value that is not a valid ISO date (agent judgement from "ISO date, empty when unpaid").	AC-3	feat.brd (BR-1, BR-2)

ID	Assumption	Used in	Source
P-A4	"Proven end to end" means the end-to-end test checks that the rendered table matches the overdue list built from the same feed (agent judgement; the BRD says only that the flow is proven end to end).	AC-22	feat.brd (S-7, SM-1)

6. Open questions (non-blocking)

6.1 Carried from the approved BRD

ID	Question	Owner (as recorded)	Source
OQ-B1	Confirm the business meaning of "vendors that usually pay late" and "chase them first" (B-A1).	PM / Leadership	feat.brd
OQ-B2	Keep "overdue" (more than 30 days) and the regulatory "Late" (more than 60 days) as distinct labels (NFR-10).	Not recorded	feat.brd
OQ-B3	Which business KPI should measure success? The inputs give none.	Not recorded	feat.brd
OQ-B4	Confirm calendar days for "days past due" in `overdue_with_risk`, with the 30-day boundary tested.	Not recorded	feat.brd
OQ-B5	Should one shared overdue definition serve this item, ZD-48501 and ZD-48211?	PM	feat.brd
OQ-B6	How should an unpaid row that has a `paid_date` be handled: reject it, or ignore the date? No acceptance criterion is written for this case until it is answered.	Not recorded	feat.brd
OQ-B7	Should an approved business question set be added to org.templates?	Not recorded	feat.brd; org.templates
OQ-B8	Should `renderOverdueRisk` HTML-escape the vendor names and ids it renders (NFR-7)?	Not recorded	feat.brd
OQ-B9	Should the portal show the risk level as visible text, or only as the row class `risk-RISK`? AC-19 follows CR-6 (id, vendor, days, amount shown) until answered.	Not recorded	feat.brd
OQ-B10	How should ap_portal format the two-decimal amount string without converting it to a binary float (NFR-3)?	Not recorded	feat.brd
—	Carried unchanged: feat.fitment FQ-3, FQ-5 and FQ-6, and the questions feat.fitment carries from feat.classification (OQ-1 to OQ-5) and feat.prebacklog (Q-1 to Q-5).	—	feat.brd (section 11)

6.2 New in this PRD

ID	Question	Why it matters	Source
PQ-1	What should `vendor_risk` return when `history` is empty? The inputs do not say how `predict_delay_days` behaves on an empty list. Agent judgement: if, under B-A3, a vendor with no paid history is given no entry in `risk_by_vendor`, its rows get `'unknown'` (AC-17); but the inputs do not say whether such a vendor gets an entry, and a direct call with an empty list is not specified.	Edge case for AC-7 to AC-9	feat.brd (BR-4, BR-11, B-A3)
PQ-2	Confirm P-A1 to P-A4 (meaning of `days`, the direction and type of the `id` tie-break, the meaning of "invalid" `paid_date`, and what the end-to-end test checks).	Exact test expectations for AC-3, AC-13, AC-16, AC-22	this file
PQ-3	Should the overdue list have a named JSON contract between ap_aging and ap_portal (field names, the type of `days`, `amount` as a string)? Proposed for the TRD (REQ-07); the type of `days` is not given in the inputs.	Interface between two coding lanes	feat.brd (BR-12, B-A7)
PQ-4	Are there performance, data volume or availability targets (NFR-11)?	Non-functional completeness	this file
PQ-5	Who decides the order of this change and PROV-ACME-2026-11, which both change `build_explanation_request`?	Delivery dependency for AC-11, AC-12	feat.brd (section 12, risks and dependencies)
PQ-6	CR-1 needs feed v3 and ZD-48470 brings feed v2. Is feed v3 available to ap_pipeline for this change?	Delivery dependency for AC-1 to AC-3	feat.brd (section 12, risks and dependencies)

7. Traceability (BRD to PRD)

BRD item	PRD coverage
S-1	US-2; AC-1, AC-2, AC-3; NFR-9
S-2	US-2; AC-4, AC-5, AC-6
S-3	US-3; AC-7, AC-8, AC-9, AC-10
S-4	US-4; AC-11, AC-12
S-5	US-1; AC-13 to AC-18
S-6	US-1; AC-19, AC-20, AC-21
S-7	US-5; AC-22, AC-23
BR-1 / BR-2	AC-1, AC-2 / AC-3
BR-3	AC-4
BR-4 / BR-5	AC-7, AC-8, AC-9
BR-6	AC-10; NFR-5
BR-7 / BR-8 / BR-9	AC-13, AC-14 / AC-15 / AC-13

BRD item	PRD coverage
BR-10 / BR-11	AC-16 / AC-17
BR-12	AC-18; NFR-1, NFR-2, NFR-3
BR-13 / BR-14	AC-19, AC-21 / AC-19, AC-20
BR-15	AC-11
BR-16	NFR-6
SM-1	AC-22
SM-2	AC-8, AC-9, AC-14; NFR-8
SM-3	AC-18; NFR-1, NFR-2, NFR-3
SM-4	AC-3
SM-5	AC-12; NFR-4
SM-6 (no business KPI)	OQ-B3 carried
OS-1 to OS-8	Summary ("Not in this change"); NFR-6, NFR-7, NFR-10
B-A1 to B-A7	Section 5.1
OQ-B6 (unpaid row with a `paid_date`)	Left open; AC-2 is limited to paid rows so that it does not decide the case

8. Validation check

Template

Check	Result	Source
PRD template sections present: summary, user stories, numbered acceptance criteria (EARS), non-functional requirements, open questions	Pass: sections 1, 2, 3, 4 and 6.	org.templates
Every acceptance criterion uses "When <trigger>, the system shall <response>"	Pass: AC-1 to AC-23.	org.templates
User stories use "As a / I want / So that"	Pass: US-1 to US-5.	org.templates

Carry-forward from the approved BRD

Check	Result	Source
Every BRD scope item S-1 to S-7 is covered	Pass (section 7).	feat.brd
Every business rule BR-1 to BR-16 is covered by an acceptance criterion or NFR	Pass (section 7).	feat.brd
Every delivery success measure SM-1 to SM-5 is covered	Pass (section 7). SM-6 has no business KPI; OQ-B3 is carried.	feat.brd
Out of scope OS-1 to OS-8 is kept, and no new scope is added	Pass. The PRD adds no function, repository or behaviour beyond S-1 to S-7.	feat.brd
BRD assumptions and open questions are carried, not silently resolved	Pass. B-A1 to B-A7 and OQ-B1 to OQ-B10 are carried. NFR-3 states a requirement for OQ-B10 as agent judgement and keeps the question open.	feat.brd
Every non-functional requirement is backed by a BRD item, a decision, a lesson or a stated source rule	Pass. Runtime versions from org.architecture are given as background only, not as a requirement.	feat.brd; record:memory; file:hobasa-agentic-sdlc/requirements/ZD-48600.md
No regulation applies, so no regulation section and page is cited	Pass, as recorded in the BRD ("No regulatory source applies to this change").	feat.brd (BQ-9)

Decision log

Decision	Result	Source
D-2026-031: aging buckets use calendar days after due date	No conflict. Applied to the overdue list and `delay_days` as assumption B-A2 (AC-4, AC-13).	record:memory; feat.brd
D-2026-044: monetary amounts use Decimal, never float	Consistent (AC-18, NFR-1 to NFR-3).	record:memory; feat.brd
D-2026-050: paid invoices are excluded from aging	Consistent (AC-15).	record:memory; feat.brd

No conflict with a recorded decision or with the approved BRD was found.

Lessons applied

Lesson	Applied in	Source
L-007: test the boundary day (exactly 30 days) explicitly for any day-count rule	AC-14, NFR-8	record:memory
L-011: keep Decimal arithmetic; never convert amounts to float	AC-18, NFR-1, NFR-3	record:memory